

UNIVERSITY OF THE PUNJAB

B.S. in Computer science Third semester : Fall
2022 Subject; Calculus and Analytical Geometry
paper: MS-251 N

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER

Q.1. Define the following: (6x5=30)

I. If $f(x) = x + 5$, $g(x) = x^2 - 3$. Find formula for $f(g(x))$. Also calculate $f(g(0))$?

Evaluate $\lim$

Find an equation for the normal line to the curve at the given point. — $xy + y^2 =$

Find the intervals of concavity and inflection points of $f(x) = x + \sin x$ for $0 \leq x \leq 2\pi$

V' Evaluate the integral $\int dx$?

VI. Find parametric equations for line through $P(-3, 2, -3)$ and $Q(1, 4)$?

Q.2. Answer the following questions. (3x10=30)

I. Find an equation for the tangent line to the curve at the given point. Then sketch the curve and tangent together. $f(x) = x^3, (-2, -8)$.

II. Find two positive numbers whose sum is 20 and whose product is as large as possible? **bf** Find all critical points of $f(x) = 4x^{1/3}$,

III. Find the point of intersection of the lines $L_1: x = 1 + 2t, y = 2 + 3t, z = 3 + 4t$

$L_2: x = 2 + t, y = 4 + 2t, z = -1 - 4t$ and then find the plane determined by these

Here's how to solve this problem
step-by-step:

1. Find the formula for $f(g(x))$

The notation $f(g(x))$ means we
substitute the function $g(x)$ into

the function $f(x)$. We are given that $f(x) = x + 5$ and $g(x) = x^2 - 3$

To find $f(g(x))$, we replace every instance of 'x' in the equation for $f(x)$ with the expression for $g(x)$:

$$f(g(x)) = (g(x)) + 5$$

Now, substitute the expression for $g(x)$:

$$f(g(x)) = (x^2 - 3) + 5$$

Simplify the expression:

$$f(g(x)) = x^2 + 2$$

Therefore, the formula for $f(g(x))$ is $x^2 + 2$.

2. Calculate $f(g(0))$

To calculate $f(g(0))$, we substitute $x = 0$ into the formula we just derived for $g(x)$

$$f(g(0)) = (0)^2 + 2$$

$$f(g(0)) = 0 + 2$$

$$f(g(0)) = 2$$

Therefore, $f(g(0)) = 2$.

Here's how to solve this limit problem:

1. Factor the numerator and denominator:

The numerator is $x^2 + x - 2$. This factors to

$$(x + 2)(x - 1).$$

The denominator is $x^2 - x$. This factors to $x(x - 1)$.

Therefore, the expression becomes:

$$\lim_{x \rightarrow 1} \frac{(x+2)(x-1)}{x(x-1)}$$

2. Simplify the expression:

Notice that $(x - 1)$ is a common factor in both the numerator and the denominator. We can cancel these factors, provided $x \neq 1$. Since we are taking the limit as x approaches 1, this condition is satisfied. The simplified expression is:

$$\lim_{x \rightarrow 1} \frac{x+2}{x}$$

3. Evaluate the limit:

Now, we can substitute $x = 1$ into the simplified expression:

1-4-2

1

Therefore, the limit is 3.

The limit of the given expression as x approaches 1 is 3.



Here's how to find the intervals of concavity and inflection points for the function $f(x) = -k \sin x$ on the interval $[0, 2\pi]$:

1. Find the first derivative:

The first derivative represents the slope of the function. We find it using the sum rule and the derivative of $\sin x$:

$$f'(x) = \frac{d}{dx}(x + \sin x) = 1 + \cos x$$

2. Find the second derivative:

The second derivative represents the rate of change of the slope, which determines concavity.

We differentiate $f'(x)$:

$$f''(x) = \frac{d}{dx}(1 + \cos x) = -\sin x$$

3. Find potential inflection points:

Inflection points occur where the second derivative changes sign. We set $f''(x) = 0$ and solve for x :

$$-\sin x = 0$$

This equation is satisfied when $x = 0, \pi, 2\pi$. These are our potential inflection points.

4. Determine concavity intervals:

We test the intervals determined by the potential inflection points to see the sign of $f''(x)$:

- Interval $(0, \pi)$: Choose a test point, say $x = \frac{\pi}{2}$. Then $f''(\frac{\pi}{2}) = -\sin(\frac{\pi}{2}) = -1 < 0$.

Thus, the function is concave down on $(0, \pi)$.

- Interval $(\pi, 2\pi)$: Choose a test point,

Here's how to find the of the

say $x = \frac{3\pi}{2}$. Then
 $f''(\frac{3\pi}{2}) = -\sin(-) = 1 > 0$.

Thus, the function is
 concave up on $(T, 2m)$.

5. Confirm inflection points:

Since the concavity changes at a:
 T , this is an inflection point. 0
 and $27T$ are endpoints and do not
 represent a change in concavity
 within the given interval.

Final Answer: The function $f(x) = x + \sin x$ is concave down on the
 interval $(0, T)$ and concave up on
 the interval $(T, 2m)$, with an
 inflection point at

equation normal line
 to the curve $a: xy + Y^2 = 7$ at the
 point $(-1, 2)$:

1. Find the derivative:

We need to find using implicit
 differentiation. Differentiate
 both sides of the equation with
 respect to x :

$$2x - (y + x \frac{dy}{dx}) + 2y \frac{dy}{dx} = 0$$

2. Solve for dx .

Rearrange the equation to isolate:

$$2x - y - x \frac{dy}{dx} + 2y \frac{dy}{dx} = 0$$

$$(2y - x) \frac{dy}{dx} = y - 2x \quad (2y - x)$$

$$\frac{dy}{dx} = \frac{y - 2x}{2y - x}$$

3. Evaluate the derivative at the
 given point:

Substitute the coordinates of the
 given point $(-1, 2)$ into the
 expression for dc :

$$\frac{dy}{dx} \big|_{(-1, 2)} = \frac{2 - 2(-1)}{2(2) - (-1)} = \frac{4}{5}$$

This is the slope of the tangent
 line at $(-1, 2)$.

Here's how to find the of the
 4. Find the slope of the normal
 line:

The normal line is perpendicular to
 the tangent line, so its slope is
 the negative reciprocal of the
 tangent line's slope:

$$m_{\text{normal}} = -\frac{1}{m} = -\frac{1}{5}$$

5. Use the point-slope form to
 find the equation of the
 normal line:

The point-slope form of a line is $y - y_1 = m(x - x_1)$
 , where (x_1, y_1) is a point on the
 line and m is the slope. Using the
 point $(-1, 2)$ and the slope
 $m = -\frac{1}{5}$

$m_{\text{normal}} = -\frac{1}{5}$ we get:

$$\begin{aligned} y - 2 &= -\frac{1}{5}(x - (-1)) \\ y - 2 &= -\frac{1}{5}(x + 1) \\ y - 2 &= -\frac{1}{5}x - \frac{1}{5} \\ y &= -\frac{1}{5}x - \frac{1}{5} + 2 \\ y &= -\frac{1}{5}x + \frac{9}{5} \end{aligned}$$

The equation of the normal line to
 the curve at the point $(-1, 2)$ is y
 $= -\frac{1}{5}x + \frac{9}{5}$.

equation normal line
 to the curve $a: xy + y^2 = 7$ at the
 point $(-1, 2)$:

1. Find the derivative:

We need to find using implicit
 differentiation. Differentiate
 both sides of the equation with
 respect to x :

$$2x - (y + x \frac{dy}{dx}) + 2y \frac{dy}{dx} = 0$$

2. Solve for dx .

Rearrange the equation to isolate:

Here's how to find the of the

$$2x - y - x \frac{dy}{dx} + 2y \frac{dy}{dx} = 0$$

$$(2y - x) \frac{dy}{dx} = y - 2x \quad (2y - x)$$

$$\frac{dy}{dx} = \frac{y - 2x}{2y - x}$$

3. Evaluate the derivative at the given point:

Substitute the coordinates of the given point $(-1, 2)$ into the expression for $\frac{dy}{dx}$:

$$\left. \frac{dy}{dx} \right|_{(-1, 2)} = \frac{2 - 2(-1)}{2(2) - (-1)} = \frac{4}{5}$$

This is the slope of the tangent line at $(-1, 2)$.

4. Find the slope of the normal line:

The normal line is perpendicular to the tangent line, so its slope is the negative reciprocal of the tangent line's slope:

$$m_{\text{normal}} = -\frac{1}{\frac{4}{5}} = -\frac{5}{4}$$

5. Use the point-slope form to find the equation of the normal line:

The point-slope form of a line is $y - y_1 = m(x - x_1)$, where (x_1, y_1) is a point on the line and m is the slope. Using the point $(-1, 2)$ and the slope

5

m

normal — we get:

$$y - 2 = -\frac{5}{4}(x - (-1))$$

$$y - 2 = -\frac{5}{4}(x + 1)$$

$$y - 2 = -\frac{5}{4}x - \frac{5}{4}$$

$$y = -\frac{5}{4}x - \frac{5}{4} + 2$$

$$y = -\frac{5}{4}x + \frac{3}{4}$$

Here's how to find the of the
The equation of the normal line to
the curve at the point $(-1, 2)$ is Y
 $= -\frac{5}{4}x + \frac{3}{4}$.



Here's how to solve both parts of the problem:

Part 1: Evaluate the integral $\int \cos^2 x \, dx$

To evaluate this integral, we use the power-reducing formula for cosine:

$$\cos^2 x = \frac{1 + \cos(2x)}{2}$$

Substitute this into the integral:

$$\int \cos^2 x \, dx = \int \frac{1 + \cos(2x)}{2} \, dx$$

Now, we can integrate term by term:

$$\begin{aligned} &= \frac{1}{2} \int (1 + \cos(2x)) \, dx \\ &= \frac{1}{2} \int 1 \, dx + \frac{1}{2} \int \cos(2x) \, dx \end{aligned}$$

The integral of 1 is x , and the integral of $\cos(2x)$ is $\frac{1}{2} \sin(2x)$.

Therefore:

$$= \frac{1}{2}x + \frac{1}{4}\sin(2x) + C$$

where C is the constant of integration.

Part 2: Find parametric equations for the line through $P(-3, 2, -3)$ and $Q(1, -1, 4)$

To find the parametric equations of a line passing through two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$, we first find the direction vector $\vec{F}$:

$$\vec{v} = \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \\ z_2 - z_1 \end{pmatrix} = \begin{pmatrix} 1 - (-3) \\ -1 - 2 \\ 4 - (-3) \end{pmatrix} = \begin{pmatrix} 4 \\ -3 \\ 7 \end{pmatrix}$$

Now, we can write the parametric equations using point P and the direction vector $\vec{F}$:

$$\begin{aligned}x &= x_1 + v_x t = -3 + 4t \\y &= y_1 + v_y t = 2 - 3t \\z &= z_1 + v_z t = -3 + 7t\end{aligned}$$

where t is a parameter.

Therefore, the parametric equations for the line are

$$x = -3 + 4t, \quad y = 2 - 3t, \quad \text{and} \quad z = -3 + 7t.$$

The integral of $\cos 2x$ is $\frac{1}{2} \sin(2x) + C$ and the parametric equations for the line are $x = -3 + 4t$, $y = 2 - 3t$, $z = -3 + 7t$.

to solve this problem step-by-step:

1. Find the derivative:

The derivative of $f(x) = x^3$ is found using the power rule: $f'(x) = 3x^2$. This represents the slope of the tangent line at any point on the curve.

2. Find the slope at the given point:

The given point is $(-2, -8)$. Substitute $x = -2$ into the derivative to find the slope of the tangent line at this point:

$$m = f'(-2) = 3(-2)^2 = 3(4) = 12$$

3. Use the point-slope form to find the equation of the tangent line:

Here's how

The point-slope form of a line is

$$y - y_1 = m(x - x_1)$$

, where (x_1, y_1) is a point on the line and m is the slope. Using the point $(-2, -8)$ and the slope $m = 12$, we get:

$$y - (-8) = 12(x - (-2))$$

$$y + 8 = 12(x + 2)$$

$$y + 8 = 12x + 24$$

$$y = 12x + 16$$

4. Sketch the curve and tangent line:

To sketch the curve $f(x) = x^3$, plot a few points. It's a cubic function that passes through the origin $(0, 0)$ and increases rapidly as x moves away from 0. The tangent line $y = 12x + 16$ has a y -intercept of 16 and a slope of 12. Draw the tangent line touching the curve at the point $(-2, -8)$.

The equation for the tangent line to the curve $y = x^3$ at the point $(-2, -8)$ is $y = 12x + 16$.

Here's how to solve these problems:

a) Finding two positive numbers

Let the two positive numbers be x and y . We are given that their sum is 20, so we can write:

$$x + y = 20$$

We want to maximize their product, $P = xy$. We can express y in terms of x using the sum equation:

$$y = 20 - x$$

Now substitute this into the product equation:

$$P(x) = x(20 - x) = 20x - x^2$$

To find the maximum, we take the derivative of $P(x)$ with respect to x and set it to zero:

$$20 - 2x = 0$$

$$2x = 20$$

Solving for x :

$$x = 10$$

$$x = 10$$

Now we find y :

$$y = 20 - x = 20 - 10 = 10$$

Therefore, the two positive numbers are 10 and 10. Their product is $10 \times 10 = 100$.

To confirm this is a maximum, we can take the second derivative:

$$\frac{d^2P}{dx^2} = -2$$

$$\frac{d^2P}{dx^2} = -2$$

Since the second derivative is negative, this confirms that we have a maximum.

The two positive numbers whose sum is 20 and whose product is as large as possible are 10 and 10.

b) Finding critical points

To find the critical points of $f(x) = x^{4/3} - 4x^{1/3}$, we need to find where the derivative is zero or undefined. First, let's find the derivative:

$$f'(x) = \frac{4}{3}x^{1/3} - \frac{4}{3}x^{-2/3}$$

Now we set the derivative equal to zero:

$$\frac{x^{1/3}}{3} - \frac{4x^{-2/3}}{3} = 0$$

$$x^{1/3} = 4x^{-2/3}$$

$$x^{1/3} = 4x^{-2/3}$$

$$x^{1/3} = 4x^{-2/3}$$

Now let's check where the derivative is undefined. The derivative is undefined when $x = 0$ because of the $x^{-2/3}$ term.

Here's how

Therefore, the critical points of $f(x)$ are $x = 0$ and $x = 1$.

The critical points of $f(x) = x^{4/3} - 4x^{1/3}$ are $x = 0$ and $x = 1$.



Here's how to solve this problem step-by-step:

1. Find the point of intersection:

To find the point of intersection of lines L_1 and L_2 , we need to find values of t and s such that the x , y , and z coordinates are equal for both lines. We set up a system of equations:

$$\begin{aligned}1 + 2t &= 2 + s \\ 2 + 3t &= 4 + 2s \\ 3 + 4t &= -1 - 4s\end{aligned}$$

We can solve this system of equations. Let's start by solving the first equation for s :

$$s = 2t - 1$$

Now substitute this into the second equation:

$$\begin{aligned}2 + 3t &= 4 + 2(2t - 1) \\ 2 + 3t &= 4 + 4t - 2 \\ 2 + 3t &= 2 + 4t \\ t &= 0\end{aligned}$$

Now substitute $t = 0$ back into the equation for s : $s = 2(0) - 1 = -1$

Finally, let's check if these values satisfy the third equation:

$$3 + 4(0) = -1 - 4(-1)$$

$$3 = 3$$

The values $t = 0$ and $s = -1$ satisfy all three equations. Now substitute $t = 0$ into the equation for L_1 (or $s = -1$ into the equation for L_2) to find the point of intersection:

$$\begin{aligned}x &= 1 + 2(0) = 1 \\ y &= 2 + 3(0) = 2 \\ z &= 3 + 4(0) = 3\end{aligned}$$

The point of intersection is $(1, 2, 3)$.

2. Find the plane determined by the lines:

To find the plane, we need two vectors that lie in the plane. we can use the direction vectors of the lines L_1 and L_2 :

$$\begin{aligned}\vec{v}_1 &= \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \quad (\text{direction vector of } L_1) \\ \vec{v}_2 &= \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \quad (\text{direction vector of } L_2)\end{aligned}$$

The normal vector to the plane is given by the cross product of these two vectors:

$$\vec{n} = \vec{v}_1 \times \vec{v}_2 = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \times \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} = \begin{pmatrix} (3)(-4) - (4)(2) \\ (4)(1) - (2)(-4) \\ (2)(2) - (3)(1) \end{pmatrix}$$

Now we use the point of intersection (1, 2, 3) and the

normal vector $\begin{pmatrix} -20 \\ 12 \\ 1 \end{pmatrix}$ to find the equation of the plane:

$$-20(x - 1) + 12(y - 2) + 1(z - 3) =$$

$$-20x + 20 + 12y - 24 + z - 3 = 0$$

$$-20x + 12y + z - 7 = 0$$

$$20x - 12y - z + 7 = 0$$

The plane determined by these lines is $-12y - z + 7 = 0$.

Final Answer: The final answer is $20x - 12y - z + 7 = 0$

